

Weekly Report: Week 4

2026-08-19

All results use the matcher `deepseek-v4-flash-0731` and the embedding model `bge-large-en-v1.5`. Every comparison within a dataset is scored over an identical candidate set: 2,665 pairs on Fodors–Zagat, 13,080 on DBLP–ACM, 88,296 on Amazon–Walmart.

1 Pipeline Overview

The pipeline runs in five stages, each timed and recorded in a per-run JSON summary under `logs/<dataset>/runs/`.

Load and normalize. Both tables and the ground truth are read, ID columns are dropped, and every remaining field is normalized: HTML entity references are decoded, text is Unicode-decomposed with combining marks stripped, and address and directional abbreviations are standardized. This matters more than it appears: `ACM.csv` stores 461 HTML entity references (`ö`, `—`) where `DBLP.csv` stores none, so without decoding, the same author name is two unrelated strings across the two sources.

Embed and block. Each attribute column is encoded independently with SentenceBERT, the column vectors are concatenated and L2-normalized into one embedding per record, and multi-probe locality-sensitive hashing retrieves the top- k most similar Table-B candidates for every Table-A record. On DBLP–ACM this reduces 6,001,104 possible pairs to 13,080 candidates while retaining all 2,224 ground-truth matches.

Step 1 — global tuple condensation (optional). One attribute subset is chosen once for the whole dataset and applied to every record. The tables may optionally be re-blocked on the condensed attributes; see Section 3.2.

Step 1.5 — pair-adaptive attribute selection (optional). A small sample of candidate pairs is profiled by the LLM for per-attribute importance, and a hybrid predictor/retriever model transfers a *different* attribute subset to every candidate pair. Blocking is unchanged, so the candidate set is identical to the no-selection baseline.

Match. One LLM call decides Yes/No per candidate pair, using only the attributes selected for that pair.

Metrics and logging. Precision, recall, and F1 are computed end-to-end, against all ground-truth matches, so pairs lost at blocking count against recall.

2 Methodology

2.1 Step 1.5: Pair-Adaptive Attribute Selection

Step 1.5 has a profiling stage and a transfer stage.

Profiling. A label-free sampler selects a small set of candidate pairs (20 on Fodors–Zagat, 50 on DBLP–ACM) and the LLM is asked, for each, which attributes matter for deciding whether the two records describe the same entity. Ground truth is not consulted at any point. The **diversity** sampler is used throughout this report, holding the sampler fixed so that other variables can be read.

Transfer. For every candidate pair, six cheap similarity features are computed per attribute — token overlap, edit similarity, exact match, semantic similarity, numeric similarity, and a numeric-type indicator — without calling the LLM. Two models then predict per-attribute importance from those features: a Ridge regressor fit on the profiled pairs, and a nearest-neighbour retriever that averages the importance vectors of the five most feature-similar profiled pairs. Their outputs are blended equally and normalized to sum to one, and a selection rule decides how many of the ranked attributes to keep.

The profiling stage costs about 1% of a run’s tokens. Its purpose is to buy a rule that can be applied to the remaining 99% without further LLM calls.

2.2 Two Ways to Elicit Importance

The profiling prompt is one of the two variables this report isolates. Two formulations were compared, identical in framing and guardrails and differing only in the answer requested.

Ordinal. The LLM assigns each attribute an integer score from 0 (not useful) to 3 (decisive), scoring each attribute independently.

Decisive. The LLM returns the smallest subset of attributes sufficient to make the decision, weighing attributes against one another in a single judgement.

The motivation for the second formulation was that the recorded 0–3 scores were not being used discriminatively. Aggregated over all runs, DBLP–ACM used score 0 on 1% of judgements, collapsing a four-point scale to three; Fodors–Zagat used score 3 on 55%. An importance vector that is nearly flat cannot express that one attribute matters more than another, whatever selection rule consumes it.

2.3 How Many Attributes to Keep

Profiling and transfer produce a ranked importance vector per pair. A separate rule turns that ranking into a subset, and it is the second variable isolated here.

Cumulative keeps attributes in descending importance until their shares sum past a fixed target, by default 0.8. It is the pipeline’s historical default and is not scale-free: importance is normalized to sum to one, so on a wide schema each attribute’s share shrinks and a fixed target becomes unreachable. On Amazon–Walmart’s 13 attributes the 0.8 target needs 7.7 attributes on average against a cap of 5, so the cap binds for every pair and the rule cannot express “this pair needs fewer fields” at all.

Gap cuts at the largest drop in the ranked importance profile. It has no threshold to tune and does not saturate on any of the three schemas.

Both rules are subject to a floor and a cap on the number of attributes kept. The floor was 2 by default and is set to 1 wherever a run is allowed to reduce a pair to a single attribute.

3 Experimental Design

3.1 Why Comparisons Are Paired

The matcher LLM is not deterministic, and on these datasets its variation is large enough to swamp the effects being measured. Running the no-selection baseline twice on Fodors–Zagat, with identical prompts and identical attributes, produced F1 of 0.8917 and 0.9596 — a gap of 0.068, from 19 of 2,665 answers changing, 18 of them from Yes to No. Precision moved from 0.823 to 0.947 while recall was unchanged.

The same measurement on DBLP–ACM gives a much smaller floor: two independent draws of the no-selection baseline produced F1 of 0.8706 and 0.8613, a gap of 0.0093. This is expected — DBLP–ACM has 2,224 ground-truth matches against Fodors–Zagat’s 110, so the same absolute number of flipped answers perturbs the metrics far less.

Comparing two methods by their F1 values from separate runs therefore measures the matcher’s variance more than it measures the methods. Two consequences follow.

First, the matcher response cache is enabled for all comparisons. Where two methods select the same attributes for a pair, the pair yields one shared answer and cannot contribute a spurious difference. Section 4.4 shows this working: across the 9,803 pairs where two very different methods happened to select the same single attribute, they recorded identical false-positive counts.

Second, methods are compared by McNemar’s test on the pairs where they actually differ, rather than by comparing aggregate F1. This is a paired test on the same candidate set and the same matcher draw, and it is the only comparison in this report that carries a significance claim.

A caveat this design imposes: because repeated runs of one configuration replay the cache, they are not independent measurements and their standard deviation is meaningless. Replication across seeds was dropped in favour of the paired design.

3.2 Holding the Candidate Set Fixed Across Methods

Step 1.5 leaves blocking untouched, so its candidate set is identical to the no-selection baseline. Step 1 condenses the record and can optionally re-block on the condensed attributes, which would change which pairs are compared. It is therefore run with re-blocking disabled (`--step1-no-reblock`) throughout, so that every configuration is scored over the same 13,080 candidate pairs containing the same 2,224 ground-truth matches, and attribute choice is the only difference between them.

4 Results

4.1 Fodors–Zagat is saturated

Method	Attrs/pair	TP	FP	FN	F1	Precision	Recall
Step 1.5, ordinal	4.00	108	6	2	0.9643	0.9474	0.9818
Step 1.5, decisive	2.68	106	4	4	0.9636	0.9636	0.9636
No selection	5.00	107	6	3	0.9596	0.9469	0.9727
Step 1, supervised	2.00	106	5	4	0.9593	0.9550	0.9636

Table 1: Fodors–Zagat, 110 ground-truth matches, `class` excluded.

Every pairwise McNemar test returns $p = 1.00$; the largest disagreement between any two methods is 9 pairs out of 2,665. Doing nothing already recovers 107 of 110 matches, so there is no headroom for a method to distinguish itself, and with 110 matches the matcher’s own instability moves F1 further than any method does. The defensible claim here is a 46% reduction in attributes at unchanged accuracy.

4.2 DBLP–ACM separates the methods

Method	Rule	Attrs/pair	TP	FP	FN	F1	Precision	Recall
Step 1.5, decisive	gap	1.42	2197	255	27	0.9397	0.8960	0.9879
Step 1, supervised (200 labels)	—	1.00	2198	339	26	0.9233	0.8664	0.9883
Step 1.5, decisive	cumulative 0.8	2.24	2214	535	10	0.8904	0.8054	0.9955
No selection	—	4.00	2216	651	8	0.8706	0.7729	0.9964
Step 1.5, ordinal	cumulative 0.8	3.77	2215	699	9	0.8622	0.7601	0.9960
Step 1, supervised (10 labels)	—	2.00	2211	739	13	0.8547	0.7495	0.9942

Table 2: DBLP–ACM, 2,224 ground-truth matches.

Comparison (A vs. B)	A right	B right	p
Decisive/gap vs. no selection	527	150	5.2×10^{-50}
Decisive/gap vs. supervised (200)	112	29	1.0×10^{-12}
Decisive/gap vs. decisive/cum. 0.8	375	112	3.5×10^{-34}
Supervised (200) vs. decisive/cum. 0.8	389	209	1.6×10^{-13}
Decisive/cum. 0.8 vs. no selection	468	350	4.2×10^{-5}
Decisive/cum. 0.8 vs. ordinal	425	258	1.7×10^{-10}
No selection vs. ordinal	135	86	1.2×10^{-3}
Ordinal vs. supervised (10)	497	453	0.16

Table 3: McNemar tests over the pairs on which two methods disagree.

The best configuration reaches F1 0.9397 on 1.42 of 4 attributes per pair, beating both no selection and a global baseline trained on 400 labelled pairs, while using no labels itself.

Recall is close to constant across the table (0.988–0.996) while precision spans 0.750–0.896. Attribute selection does not help the matcher find matches it was missing; it stops the matcher asserting matches that are not there. The one cost appears at the aggressive end, where the configurations showing barely more than one attribute give up 18–19 of 2,224 matches.

Holding the rule fixed, the decisive formulation beats no selection ($p = 4.2 \times 10^{-5}$) while the ordinal formulation is significantly *worse* than no selection ($p = 1.2 \times 10^{-3}$). Only the profiling question differs between them. Ordinal also barely selects, keeping all four attributes on 76.8% of pairs. The attribute decisive learns to discard is **venue**: the two sources render it incompatibly ("SIGMOD Record" against "The VLDB Journal --- The International Journal on..."), so venue disagreement is evidence of formatting, not of non-identity. Scored in isolation it looks moderately informative, which is why ordinal keeps it.

4.3 The selection rule matters as much as the elicitation

At the default target of 0.8, pair-adaptive selection *loses* to the global baseline ($p = 1.6 \times 10^{-13}$); at 0.4 it wins (66/2, $p = 1.6 \times 10^{-17}$). The comparison between adaptive and global selection turned

Rule	Attrs/pair	FP	FN	F1	<code>title</code> only
cumulative 0.8 (default)	2.24	535	10	0.8904	0.0%
cumulative 0.6	1.66	434	13	0.9082	34.5%
cumulative 0.5	1.27	293	20	0.9337	73.2%
cumulative 0.4	1.18	275	26	0.9359	82.7%
gap (no threshold)	1.42	255	27	0.9397	74.9%
Global baseline, fixed <code>[title]</code>	1.00	339	26	0.9233	100.0%

Table 4: DBLP-ACM, decisive profiling throughout, varying only the selection rule.

on an untuned hyperparameter rather than on the methods.

The best configuration is at neither extreme: showing everything (4.00 attributes) and showing one fixed attribute (1.00) are both beaten by rules showing roughly 1.2–1.4. Targets 0.4 and 0.5 are statistically indistinguishable (55/67, $p = 0.32$), so this is a plateau rather than one fortunate value, while the steps above them are individually significant. The `gap` rule is the practically important row: it has no threshold, so nothing about it was chosen by inspecting these results, and it reaches the top of the range anyway. Its margin over the best hand-tuned target is thin (50/31, $p = 0.045$) and should not be claimed as superiority — the defensible claim is that it matches a tuned threshold without requiring one.

4.4 Why adaptation beats a fixed subset

The global baseline reduces every record to `title`, which is also what the adaptive method chooses for three quarters of pairs. Comparing them isolates adaptation itself, holding the attribute vocabulary fixed.

Pair group	Pairs	FP, adaptive	FP, fixed <code>[title]</code>
Adaptive also chose <code>[title]</code>	9,803	118	118
Adaptive chose more	3,277	137	221
<code>[title, authors, year]</code>	2,257	101	146
<code>[title, year]</code>	841	13	58
<code>[title, authors]</code>	179	23	17

Table 5: DBLP-ACM, all 13,080 pairs: decisive under `gap` against the 200-label global baseline.

Where the two select the same attribute they produce identical results — 118 false positives each — which is what the shared cache guarantees and which confirms the comparison is clean. The entire gain comes from the 3,277 pairs where the adaptive method kept more, cutting false positives from 221 to 137. The clearest single effect is the 841 pairs given `[title, year]`, where false positives fall from 58 to 13: titles similar enough to mislead a title-only matcher, with the year settling the question. A global method cannot express this, because adding `year` everywhere is close to what the cumulative-0.8 configuration does, and it costs more elsewhere than it saves here.

Method	Attrs/pair	TP	FP	FN	F1	Precision	Recall
No selection (full)	12.00	1010	1018	35	0.6573	0.4980	0.9665
Step 1.5, decisive (gap)	—	—	—	—	—	—	—
Step 1, supervised	—	—	—	—	—	—	—
Step 1.5, ordinal (gap)	—	—	—	—	—	—	—

Table 6: Amazon–Walmart, 1,143 ground-truth matches, blocking completeness 0.9143. The three selection configurations were still running.

4.5 Amazon–Walmart: baseline only

This is the benchmark with headroom. Baseline precision is 0.4980 here against 0.7729 on DBLP–ACM and 0.9469 on Fodors–Zagat — close to half the asserted matches are wrong. Recall is 0.9665 against a blocking ceiling of 0.9143, so the matcher recovers nearly every candidate available to it and the whole deficit is in precision, the quantity attribute selection was shown to improve in Section 4.2. It is also where condensation is worth most: 1,035 prompt tokens per pair at the full representation, against 306 on DBLP–ACM.

One structural finding came out of preparing this dataset. The `original_id` field is a per-table row counter running 1...22,074 and 1...2,554, sharing no key across the tables. Excluding it raised pair completeness from 0.9099 to 0.9143, recovering five ground-truth pairs. Because each column is encoded independently and the column vectors are concatenated, that field occupied a full block of every record embedding and displaced real matches out of the candidate set. No attribute selection could have recovered them, since Step 1.5 does not re-block — the repair had to happen before embedding.

5 Discussion

Pair-adaptive selection outperforms both no selection and a labelled global baseline.

On DBLP–ACM it improves F1 from 0.8706 to 0.9397 while showing the matcher 65% fewer attributes ($p = 5.2 \times 10^{-50}$), and beats a baseline trained on 400 labelled pairs over the identical candidate set ($p = 1.0 \times 10^{-12}$) while using none.

How importance is elicited decides whether adaptation helps at all. The ordinal formulation, on the same pipeline and rule, is significantly worse than no selection. Scoring each attribute in isolation produces a nearly flat importance vector that no rule can act on; asking for a sufficient subset forces the comparison the task requires. This is the most transferable finding here, and it is a property of the elicitation rather than of the dataset.

The rule consuming the importance vector is as consequential as the vector. The same profiling output, read by the default rule, loses to the global baseline; read by `gap`, it beats it. Any claim about pair-adaptive selection that does not state its selection rule is under-specified. The default was not merely suboptimal but, on wide schemas, structurally incapable of condensing at all.

The benefit is conditional on the benchmark having headroom. On Fodors–Zagat no method differs from any other. DBLP–ACM has headroom and rewards adaptation, but `title` nearly identifies a paper on its own, which compresses the winnable margin to 141 discordant pairs in 13,080. Amazon–Walmart, at baseline precision 0.4980 and with no such key, is the sharper test.

6 Limitations

Amazon–Walmart is incomplete. Only its baseline is measured. The central claim therefore rests on DBLP–ACM alone, a dataset with a near-unique key. This is the main gap.

The selection rule was chosen on DBLP–ACM; its confirmation is outstanding. The sweep in Section 4.3 was read on the same pairs it reports over, so the cumulative targets in it are tuned values and none should be quoted as the method’s performance. The rule adopted, `gap`, has no threshold and so contributes no tuned quantity, but preferring it over `cumulative` was itself a choice made after seeing those results.

The design handles this by role separation: DBLP–ACM is the *selection* dataset and Amazon–Walmart the *held-out* one. `gap` was fixed before any Amazon–Walmart selection run started, and no Amazon–Walmart result will be used to revise it, so those runs are an out-of-sample test rather than a further chance to tune. The open issue is therefore not that the choice is unjustifiable but that its confirmation is not yet in: only the Amazon–Walmart baseline has completed.

Single matcher draw. Each configuration was run once. The paired design makes the comparisons valid within that draw, but does not establish that effect sizes replicate under a different one. The large DBLP–ACM effects are unlikely to reverse; the `gap`-versus-tuned-threshold comparison, with 81 discordant pairs and $p = 0.045$, could.

The supervised baseline is not an equal competitor. It consumes 400 labelled pairs while the adaptive method consumes none, so the comparison measures method and label budget together. Its selected subset also moves with the budget — `[title, authors]` at 10 and 50 labels, `[title]` at 200 — so no single sample size characterises it.

The feature vector is weaker than it appears. Of the six per-attribute features, `semantic_similarity` is disabled and constant at zero and `is_numeric` is constant per column on both datasets reported here; on the widest schema `exact_match` fires on 0.06% of cells. Effectively three features carry signal. The vector also has no notion of value rarity, the standard discriminative signal in entity resolution: matching on a rare title token and on a common one are represented identically.

One sampler. All Step 1.5 runs use the `diversity` profile sampler, held fixed so the other variables could be read. No claim about sampler choice is made.